

Model Lab

Every prior, same slate — 2026-09-23

SERVED PRIOR

Core (KNN+SVR)

PRIORS COMPARED

10

HELD-OUT PA

37,465

Held-Out Scores

Fitted on plate appearances before 2026-08-16, scored on the 37,465 that came after (actual HR rate 2.97%). No outcome in the test window was visible during fitting.

Metrics are per plate appearance; the probabilities lower down are per game.

Read AUC first — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5706	1.51×	0.02880	0.13302	0.00284
KNN comps Prior is the pooled rate of the nearest comparables.	0.5855	1.49×	0.02882	0.13294	0.00533
SVR Support-vector regression on the same feature matrix.	0.5799	1.39×	0.02889	0.13427	0.00854
Random forest Forest regression; a learned similarity metric.	0.5786	1.43×	0.02897	0.13434	0.00914
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5855	1.49×	0.02882	0.13294	0.00533
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5851	1.48×	0.02882	0.13297	0.00552
Linear regression Ordinary least squares on the same features.	0.5797	1.27×	0.02898	0.13480	0.01045
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5835	1.41×	0.02897	0.13398	0.00866
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5837	1.33×	0.02891	0.13362	0.00843
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5817	1.48×	0.02893	0.13410	0.00843

Blend weights (forest variant): KNN 0.95, SVR 0.00, RF 0.05

Shrinkage K per prior: league 249, knn 249, svr 249, rf 249, core 249, ensemble 249, linear 249, logistic 249, nn 249, xgb 249 — estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-Sample Subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (4,384 PA, actual HR rate 2.35%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5258	1.17×	0.02299	0.11211	0.01116
KNN comps Prior is the pooled rate of the nearest comparables.	0.5336	0.78×	0.02329	0.11634	0.01959
SVR Support-vector regression on the same feature matrix.	0.5563	0.97×	0.02358	0.12137	0.02538
Random forest Forest regression; a learned similarity metric.	0.5454	0.58×	0.02437	0.12358	0.03826
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5336	0.78×	0.02329	0.11634	0.01959
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5328	0.78×	0.02333	0.11643	0.02064
Linear regression Ordinary least squares on the same features.	0.5353	0.58×	0.02451	0.12761	0.04017
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5358	0.58×	0.02440	0.12290	0.03853
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5463	0.78×	0.02389	0.11968	0.03119
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5471	1.07×	0.02401	0.12235	0.03185

Slate Agreement

Flat league · 4/6 shared 0/4 hit

Miguel Vargas
Pete Crow-Armstrong
Fernando Tatis Jr.
Kyle Schwarber
Zach Neto
Luis García Jr.

SVR · 5/6 shared 2/6 hit

Miguel Vargas
Kyle Schwarber
Ben Rice ✓
Juan Soto ✓
Luis García Jr.
Pete Crow-Armstrong

Core (KNN+SVR) · 6/6 shared 2/5 hit

Miguel Vargas
Kyle Schwarber
Zach Neto
Pete Crow-Armstrong
Ben Rice ✓
Juan Soto ✓

Linear regression · 4/6 shared 2/5 hit

Miguel Vargas
Kyle Schwarber
Jose Siri
Emmanuel Rodriguez
Juan Soto ✓
Ben Rice ✓

Neural net (torch) · 5/6 shared 2/5 hit

KNN comps · 6/6 shared 2/5 hit

15.9%	Miguel Vargas	19.5%
15.7%	Kyle Schwarber	17.8%
15.6%	Zach Neto	17.5%
15.4%	Pete Crow-Armstrong	17.3%
15.3%	Ben Rice ✓	17.3%
15.2%	Juan Soto ✓	17.1%

Random forest · 4/6 shared 1/5 hit

19.4%	Miguel Vargas	19.1%
19.3%	Kyle Schwarber	18.9%
18.6%	Emmanuel Rodriguez	18.1%
18.6%	Ben Rice ✓	18.0%
18.1%	Lazaro Montes	18.0%
18.1%	Pete Crow-Armstrong	17.8%

Forest blend · 6/6 shared 2/5 hit

19.5%	Miguel Vargas	19.5%
17.8%	Kyle Schwarber	17.9%
17.5%	Zach Neto	17.5%
17.3%	Pete Crow-Armstrong	17.4%
17.3%	Ben Rice ✓	17.4%
17.1%	Juan Soto ✓	17.1%

Logistic regression · 3/6 shared 0/4 hit

20.0%	Kyle Schwarber	20.9%
18.6%	Miguel Vargas	20.1%
18.1%	Emmanuel Rodriguez	19.0%
17.7%	Pete Crow-Armstrong	18.6%
17.7%	Yordan Alvarez	18.3%
17.7%	Cal Raleigh	17.9%

Jose Siri

Pete Crow-Armstrong

17.7%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
1. Miguel Vargas slate Chicago White Sox · vs LHP · 33 HR / 685 PA	15.9%	19.5%	19.4%	19.1%	19.5%	19.5%	20.0%	20.1%	20.1%
2. Kyle Schwarber Philadelphia Phillies · vs RHP · 45 HR / 678 PA	15.4%	17.8%	19.3%	18.9%	17.8%	17.9%	18.6%	20.9%	19.1%
3. Zach Neto slate Los Angeles Angels · vs RHP · 26 HR / 680 PA	15.3%	17.5%	17.5%	17.3%	17.5%	17.5%	16.7%	17.7%	16.7%
4. Pete Crow-Armstrong slate Chicago Cubs · vs RHP · 45 HR / 708 PA	15.7%	17.3%	18.1%	17.8%	17.3%	17.4%	17.4%	18.6%	17.9%
5. Ben Rice slate New York Yankees · vs RHP · 39 HR / 649 PA	15.2%	17.3%	18.6%	18.0%	17.3%	17.4%	17.7%	17.9%	18.1%
6. Juan Soto slate New York Mets · vs RHP · 26 HR / 458 PA	14.1%	17.1%	18.6%	17.3%	17.1%	17.1%	17.7%	17.5%	17.9%
7. Jake Bauers Milwaukee Brewers · vs RHP · 26 HR / 556 PA	13.9%	16.6%	16.3%	16.7%	16.6%	16.6%	15.8%	15.9%	16.4%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
8. Luis García Jr. New York Yankees · vs RHP · 30 HR / 534 PA	15.2%	16.6%	18.1%	17.6%	16.6%	16.6%	17.1%	17.6%	17.5%
9. Mickey Moniak Colorado Rockies · vs RHP · 23 HR / 429 PA	14.4%	16.6%	17.7%	17.0%	16.6%	16.6%	16.3%	17.2%	17.0%
10. Lazaro Montes Seattle Mariners · vs RHP · 4 HR / 46 PA	12.8%	16.5%	14.3%	18.0%	16.5%	16.6%	17.5%	16.4%	16.5%
11. Jose Siri Los Angeles Angels · vs RHP · 9 HR / 240 PA	14.1%	16.5%	17.9%	17.2%	16.5%	16.5%	18.1%	17.8%	17.8%
12. Pete Alonso slate Baltimore Orioles · vs RHP · 40 HR / 682 PA	14.6%	16.5%	16.7%	16.7%	16.5%	16.5%	16.7%	17.1%	16.8%
13. Fernando Tatis Jr. San Diego Padres · vs LHP · 25 HR / 687 PA	15.6%	16.5%	16.0%	15.3%	16.5%	16.4%	16.9%	17.2%	16.6%
14. Yordan Alvarez Houston Astros · vs RHP · 41 HR / 667 PA	14.0%	16.4%	17.9%	17.3%	16.4%	16.5%	17.1%	18.3%	17.7%
15. Heriberto Hernández Miami Marlins · vs RHP · 24 HR / 485 PA	13.7%	16.1%	16.1%	16.1%	16.1%	16.1%	16.0%	16.1%	16.3%
16. Mike Trout Los Angeles Angels · vs RHP · 21 HR / 615 PA	13.4%	16.0%	15.4%	16.0%	16.0%	16.0%	16.1%	16.8%	16.4%
17. Cal Raleigh Seattle Mariners · vs RHP · 23 HR / 511 PA	13.7%	16.0%	15.1%	16.2%	16.0%	16.0%	16.8%	17.9%	16.6%
18. Pete Alonso slate Baltimore Orioles · vs RHP · 40 HR / 682 PA	14.1%	15.9%	16.2%	16.2%	15.9%	16.0%	16.1%	16.5%	16.3%
19. Griffin Conine Miami Marlins · vs RHP · 17 HR / 306 PA	13.4%	15.9%	17.1%	16.4%	15.9%	15.9%	16.7%	17.0%	16.9%
20. Kyle Stowers Miami Marlins · vs RHP · 20 HR / 507 PA	13.0%	15.9%	16.1%	15.4%	15.9%	15.8%	14.8%	15.9%	15.7%
21. Shohei Ohtani Los Angeles Dodgers · vs RHP · 30 HR / 596 PA	13.4%	15.9%	16.6%	16.5%	15.9%	15.9%	16.2%	17.0%	16.7%
22. Hunter Goodman	14.5%	15.7%	16.8%	16.1%	15.7%	15.7%	15.7%	16.6%	16.5%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)
37. Bryce Harper Philadelphia Phillies · vs RHP · 28 HR / 671 PA	13.3%	14.6%	14.9%	15.2%	14.6%	14.7%	14.5%	14.8%	14.8%
38. Jonathan Aranda Tampa Bay Rays · vs RHP · 22 HR / 652 PA	12.8%	14.6%	13.6%	14.0%	14.6%	14.6%	14.1%	13.7%	13.8%
39. William Contreras Milwaukee Brewers · vs RHP · 18 HR / 624 PA	14.2%	14.6%	14.6%	13.9%	14.6%	14.5%	14.1%	14.3%	14.3%
40. Ty France San Diego Padres · vs LHP · 21 HR / 478 PA	13.0%	14.6%	14.3%	14.0%	14.6%	14.5%	14.2%	14.1%	14.1%

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context.